AP CYBERSECURITY | INVESTIGATION LAB - STUDENT

LAB 2

Who Entered the Robotics Lab?

Unit 2: Securing Spaces | 45 to 60 minutes

Your role

You are the school's security analyst. After Friday's class, the robotics teacher found equipment missing from room R-114. You must work out **how someone could have gotten in, what the evidence proves, what it does not prove**, and which physical controls should be added.

What you will do

- Combine badge, camera, visitor, and inventory evidence into one timeline
- Spot tailgating by comparing badge entries with people on camera
- Separate what the evidence proves from what it only suggests
- Identify physical vulnerabilities and choose preventive, detective, and deterrent controls
- Rate the risk and recommend a layered fix

EXAM LENS
Every lab follows the same four stages: **Learn the concept, Investigate the evidence, Make a security decision, then Answer AP-style questions.** Commands show you what happened. The skill is deciding what the evidence *proves*, what it *does not prove*, and what to do next.

COMMON TRAP
This is a **fictional, controlled** investigation. Never scan, test, or access real school systems, other people's accounts, or any device you do not own or have written permission to use.

Original practice material written in the style of the AP Cybersecurity exam. Not reproduced from College Board.

Mission Briefing

At 4:05 PM on Friday, Oct 9, Ms. Okafor audited the robotics storage and found missing items. Four sources of evidence exist, and none of them tells the whole story. Your job is to combine them carefully and avoid blaming anyone without proof.

Evidence files

File	What it is
badge_log.csv	Badge-reader log for door R-114 (entries and denied attempts)
camera_notes.txt	Notes from reviewing the hallway camera (Camera 4)
visitor_log.csv	Front-office visitor sign-in and sign-out sheet
inventory.csv	Equipment counts at 1:30 PM and 4:05 PM

Facts about the space

- The R-114 door has a badge reader on the **outside only**. People leave by pushing a bar, which is not logged.
- There is a camera in the hallway pointed at the door. There is **no camera inside the lab**.
- Visitor badges (V-series) are **not** programmed to open R-114.
- A door-held-open alarm records an event but does not call anyone.

STAGE 1 Learn

Be able to explain these before you touch the evidence

- ■ I can define tailgating and name two controls that prevent it.
- ■ I can sort controls into deterrent, preventive, and detective.
- ■ I can explain why a log can be accurate and still miss what happened.
- ■ I can use likelihood and impact to rate a risk.

STAGE 2 Investigate

Collect facts. Record what you observe, not what you assume.

1. Build a **combined timeline** for 3:39 PM to 4:05 PM. For each event, write the time, what happened, and which file it came from.

2. At 3:41 PM, how many badge entries does the badge log show, and how many people does the camera show entering? What does the difference mean?

3. Badge B3318 (Tyler R.) was denied three times. List the **facts**, then list what you **cannot conclude**.

```
2026-10-09 14:40:55,R-114,B3318,Tyler R. (student),DENIED - badge expired 2026-09-30
2026-10-09 14:41:09,R-114,B3318,Tyler R. (student),DENIED - badge expired 2026-09-30
2026-10-09 15:39:07,R-114,B3318,Tyler R. (student),DENIED - badge expired 2026-09-30
```

4. What does the 3:42:30 alarm entry tell you? Was it a **preventive**, **detective**, or **deterrent** control? Is there any evidence anyone responded?

5. Use `inventory.csv`. What is missing? Over what time window could it have been taken?

6. Which of the four files has **no information about people leaving** R-114? Why does that matter for an investigation?

7. Look at the visitor log. D. Kim signed in at 3:35 PM and never signed out, and badge V-07 was not returned. Does this prove Kim took the equipment? What would you do next?

```
2026-10-09,R. Alvarez (parent),14:50,Counselor office,15:20,V-03 returned
2026-10-09,D. Kim (vendor),15:35,Printer service - main office,,V-07 NOT returned
```

8. List at least **four physical or procedural vulnerabilities** this incident exposed.

9. Complete the table: choose four controls, say whether each is **preventive, detective, or deterrent**, and name the vulnerability it addresses.

10. Rate the **risk** to the robotics equipment before the fix (low, moderate, high). Explain with likelihood and impact.

STAGE 3 Make a Security Decision

Turn evidence into a defensible recommendation

What can you confirm? (at least three facts)

What is not proven? (at least two limits of the evidence)

What do you recommend? Give a layered plan: one preventive, one detective, and one administrative step.

STAGE 4 AP-Style Questions

Transfer what you learned to a new situation

1. At 3:41 PM, one badge tap opens R-114 and the camera shows two people entering.

Which physical security failure does this describe?

- **A.** Shoulder surfing
- **B.** Tailgating
- **C.** Baiting
- **D.** Dumpster diving

2. The school wants to prevent unauthorized people from following staff into R-114, where staff often carry equipment with both hands.

Which control would best accomplish this?

- **A.** A camera above the door
- **B.** A sign reading 'No Tailgating'
- **C.** An access control vestibule
- **D.** A visitor log at the front office

3. An analyst reviews the evidence from R-114.

Which conclusion is best supported by the evidence?

- **A.** Tyler R. stole the kits because his badge was denied
- **B.** Lena P. stole the kits because her badge opened the door
- **C.** D. Kim entered the lab using visitor badge V-07
- **D.** An unidentified person entered R-114 without a badge tap by following Lena P.

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4. The badge system records an alarm when a door is held open for more than 60 seconds, but it does not notify anyone.

Which type of control is this alarm entry?

A. Detective
B. Preventive
C. Deterrent
D. Corrective

Evidence Printout

Printed copies of the evidence files, for students without a terminal. The files themselves are in the evidence folder.

badge_log.csv

```
time,door,badge_id,holder,event
2026-10-09 13:25:10,R-114,B1042,Ms. Okafor (teacher),GRANTED
2026-10-09 13:31:44,R-114,B2207,Lena P. (student),GRANTED
2026-10-09 13:32:50,R-114,B2209,Marcus D. (student),GRANTED
2026-10-09 14:40:55,R-114,B3318,Tyler R. (student),DENIED - badge expired 2026-09-30
2026-10-09 14:41:09,R-114,B3318,Tyler R. (student),DENIED - badge expired 2026-09-30
2026-10-09 15:10:20,R-114,B1042,Ms. Okafor (teacher),GRANTED
2026-10-09 15:39:07,R-114,B3318,Tyler R. (student),DENIED - badge expired 2026-09-30
2026-10-09 15:41:02,R-114,B2207,Lena P. (student),GRANTED
2026-10-09 15:42:30,R-114,,,ALARM - door held open 74 seconds
2026-10-09 16:05:00,R-114,B1042,Ms. Okafor (teacher),GRANTED
```

camera_notes.txt

```
CAMERA REVIEW NOTES - prepared by Mr. Hale (front office) from Camera 4, hallway outside R-114
Note: There is NO camera inside the robotics lab. Camera 4 shows only the hallway and the door.

15:39:10  A person in a gray hooded sweatshirt stands near R-114. No lanyard visible. Face not visible.
15:41:00  Lena P. taps her badge. Door opens. The gray-hoodie person steps in directly behind her before
    the door closes.
15:42:30  A cardboard box is leaning against the open door. Door stays open about 74 seconds.
15:58:30  A gray-hoodie person leaves R-114 by the push bar (exit does not need a badge). A backpack
    looks fuller.
15:58:50  Person walks toward the east stairwell. Camera 4 view ends.
```

visitor_log.csv

```
date,visitor,sign_in,visiting,sign_out,visitor_badge
2026-10-09,R. Alvarez (parent),14:50,Counselor office,15:20,V-03 returned
2026-10-09,D. Kim (vendor),15:35,Printer service - main office,,V-07 NOT returned
```

inventory.csv

```
item,count_at_13:30,count_at_16:05
Arduino kit,6,3
Laptop,2,2
Servo pack,4,3
3D printer filament,5,5
```

KEY IDEA

Before you turn this in: every answer should name the evidence it comes from (a file name, a line, a time, or a command result). If you are not sure something is proven, write 'not confirmed' and say what evidence would confirm it.